

Ata Utku Özkan

PhD Candidate, Computational Physics & Stochastic Modelling

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Professional Summary

- PhD researcher (Bilkent UNAM, exp. 2026) with 6+ years building numerical simulation and statistical inference pipelines at scale; published work covers transition-rate estimation from time-series data, regime-switching transport modelling, and phase-diagram construction over continuous parameter spaces, executed across 100+ independent runs on national HPC with fully automated Python analysis.
- Production-grade Python and C++ developer: stochastic modelling and fitting (SciPy, statsmodels), ML (PyTorch, scikit-learn), time-series and heavy-tail statistics (ARMA/GARCH, EVT), large-scale parallel computation (MPI, SLURM); open-source finance_toolkit covering tail-risk functions and heavy-tail distribution fitting (github.com/Ata-gith).
- Strong foundations in probability theory, numerical methods, and high-performance computation; seeking to apply rigorous quantitative modelling in a research or development role in systematic trading, risk, or derivatives.

Technical Skills

Programming & Data	Python (NumPy, pandas, SciPy, statsmodels, scikit-learn, PyTorch, TensorFlow/TF-Probability), C/C++ (multithreading, MPI), R, SQL (MySQL, SQLite), Bash, LaTeX, Git, Linux/HPC (SLURM)
Stochastic Modelling	Stochastic differential equations (Ornstein-Uhlenbeck, geometric Brownian motion, jump-diffusion, Lévy processes); transition-rate estimation and free-energy calibration from time-series data; regime-switching and non-linear transport modelling; phase-diagram construction over continuous parameter spaces
Statistical Methods	Time-series modelling (ARMA, GARCH); extreme value theory and heavy-tail distributions; Bayesian inference and parameter estimation; Monte Carlo simulation (importance sampling, variance reduction, path-dependent pricing); numerical PDE solvers (finite differences, finite elements); hypothesis testing and cross-validation
Machine Learning	Supervised and unsupervised learning; neural networks; gradient boosting (XGBoost, LightGBM); graph neural networks (PyTorch Geometric); dimensionality reduction (PCA, UMAP); uncertainty quantification
High-Performance Computing	MPI parallelisation; SLURM job scheduling; GPU acceleration (CUDA via PyTorch); large-scale batch computation (100+ concurrent jobs); performance optimisation in C++/Python
Languages	English (advanced), Turkish (native); German and French (A2)

Research & Professional Experience

2022 – present **PhD Researcher** — Bilkent University, UNAM, Ankara
Supervisor: Prof. Aykut Erbaş – Dept. Materials Science and Nanotechnology

▷ **Non-equilibrium transport and regime detection (first author, under review *Phys. Rev. Mater.*)** Ran 100+ independent simulations (up to 100 ns each) on HPC to study how a continuously tunable control parameter drives a confined system between distinct transport regimes. Extracted steady-state drift rates by fitting a two-stage relaxation model to time-series displacement data; showed that response is non-linear and non-monotonic, with a phase boundary between single-particle and collective transport that depends on the interaction parameter ratio. Identified the microscopic mechanism behind experimentally observed hysteretic current-voltage behaviour.

▷ **Phase-diagram construction and network topology analysis (second author, *bioRxiv* 2026)** Built end-to-end Python analysis pipeline for a large-scale polymer simulation (1250-chain system): graph reconstruction from particle coordinates, vacancy-filling algorithm for gap-size distribution, surface-coverage metrics, and edge/face network statistics. Mapped a 2D phase diagram over two continuous parameters, identified transition boundaries, and validated simulated topology against experimental imaging data.

▷ All simulation batches scheduled via SLURM on national HPC; fitting, analysis, and visualisation fully automated in Python (NumPy, SciPy, pandas, NetworkX).

2019 – 2022 **MS Researcher** — Bilkent University, UNAM, Ankara

Thesis: Molecular Dynamics Simulations Study of Rotaxane in Bulk and at Interfaces

▷ **Transition-rate estimation and free-energy calibration from time-series data (first author, *JPCB* 2023)** Measured switching timescales across 18 distinct system states using 20 independent replicas per transition; fitted exponential relaxation models to time-series observables to extract forward and reverse rates ($\tau = 0.9$ and 3.7 ns); back-calculated free-energy barriers from transition-state theory. Demonstrated robustness by replicating results under two independent model parametrisations. The pipeline — fit a relaxation model, extract a rate, infer an underlying energy landscape — is structurally identical to calibrating a mean-reversion or diffusion model from observed price paths.

▷ Characterised how a single continuous control parameter (molecular charge state) tunes equilibrium structure, solvent exposure, and network porosity across a 2D interfacial system; showed that small parameter perturbations produce large, measurable shifts in observable distributions.

Mar – Jun 2017 **Computational Risk Analysis Intern** — Nurol Makine ve Sanayi A.Ş.

Statistical forecasting and risk modelling in quality control; built regression models for defect-rate prediction from time-series production data.

Jan – Mar 2017 **Research Intern** — METU Nanomagnetism Instruments

Quantitative surface analysis; noise characterisation and signal processing on instrument time-series data.

Jan – Mar 2016 **Engineering Intern** — Aydokum A.Ş.

Finite-element simulation; numerical analysis of microstructure phase data.

Education

2022 – present

■ **PhD, Materials Science and Nanotechnology** — Bilkent University

National Nanotechnology Research Center (UNAM), Ankara

Dissertation: Stochastic Transport and Self-Assembly in Soft and 2D Materials — numerical methods: Langevin dynamics, Fokker-Planck, Monte Carlo, Bayesian inference

Education (continued)

- 2019 – 2022  **M.Sc., Materials Science and Nanotechnology** — Bilkent University
National Nanotechnology Research Center (UNAM), Ankara
Thesis: Molecular Dynamics Simulations Study of Rotaxane in Bulk and at Interfaces — methods: thermodynamic integration, free-energy estimation, statistical inference
- 2014 – 2019  **B.S., Materials Science and Nanotechnology Engineering** — TOBB ETU, Ankara
Thesis: Computational Modeling of Peptide–Graphene Interactions

Fellowships & Scholarships

- 2025  **TÜBİTAK 2211-C** – National PhD Scholarship in Priority Fields in Science and Technology
- 2024  **TÜBİTAK 2224-A** – Grant for Participation in Scientific Meetings Abroad
-  **TÜBİTAK 2250** – Performance-Based Scholarship for PhD Scholars

Talks & Certifications

Invited and Contributed Talks

- 2024  **CCP5 Annual General Meeting 2024**
Stochastic Ionic Transport and Diffusion in 2D Layered Materials: Langevin Modelling and Monte Carlo Benchmarking
agm44.ccp5.ac.uk
-  **14th Chemical Physics Congress**
Molecular Dynamics and Statistical Inference for Ionic Conductivity in 2D-MnO Systems
https://cpc14.trakya.edu.tr

Certifications

- 2022  **Machine Learning** — Stanford Online / Coursera
https://www.coursera.org/account/accomplishments/verify/T65HY2KGSJLZ
-  **Multithreading with Modern C++; ML and Data Science Bootcamp; Advanced Modern C++** — Udemy
- 2021  **Object-Oriented Data Structures in C++** — University of Illinois Urbana-Champaign / Coursera
https://www.coursera.org/account/accomplishments/verify/HR9HNA5EP5UK

Publications

Peer-Reviewed Journal Articles

- 1  **A. U. Özkan**, D. Tuncel, and A. Erbaş, “Effect of charge state on the equilibrium and kinetic properties of mechanically interlocked [5]Rotaxane: A molecular dynamics study,” *The Journal of Physical Chemistry B*, vol. 127, no. 5, pp. 1254–1263, 2023, PMID: 36716388.  DOI: 10.1021/acs.jpcc.2c07645.

Manuscripts Under Review

- 1  **A. U. Özkan**, T. S. Kasırga, and A. Erbaş, “Ionic transport in spontaneously ion-intercalated van der Waals layered structures,” 2026, Manuscript under review. arXiv: 2601.14836 [cond-mat.mtrl-sci].  URL: https://arxiv.org/abs/2601.14836.

Preprints

- 1  H. A. Hameed, **A. U. Ozkan**, and A. Erbas, “Inter-lamin interactions control meshwork topology in a polymer-gel model of nuclear lamina,” *bioRxiv*, 2026, Preprint.  DOI: 10.64898/2026.03.14.711786.

References

Available on Request